Data Communication and Computer Networks

Chapter 4 | Network and Transport Layer

Unit 4 - Addressing, IP protocols, routing, TCP, UDP and congestion control (6 hrs)

Syllabus covered

Host-to-host delivery, internetworking, IPv4 addressing and subnetting, network layer protocols (ARP, IPv4, ICMP, IPv6); transport layer - process-to-process delivery, UDP, TCP, congestion control; routing protocols - AODV, DSDV, DSR.

What is inside

Section	Question type	Marks	Count
Section A	Short answer - definitions, one-liners, small numericals	2	34
Section B	Long answer - full explanations with tables and diagrams	7	12

How to use this bank

- Subnetting is the highest-scoring topic in this chapter. Learn the five-step routine in Section B and apply it to every address you meet.
- Memorise the header field lists of IPv4, IPv6 and TCP as sequences, not sentences.
- Keep TCP and UDP in one comparison table in your head - almost every paper asks for it.
- For congestion control remember the split: open loop prevents, closed loop removes.

Source tags

- **PYQ <year>** - asked in a C. V. Raman Global University end-semester paper (2022, 2023, 2024-25 or the 2024-25 backlog paper).
- **Model** - from the printed unit-wise model question paper.
- **Assignment** - from the course assignment for this unit.
- **Notes** - from the lecture slides, not yet asked but examinable.

Section A | 2-mark questions

Answer in two to four lines. Each point below is one mark-bearing idea, so a single line written in the exam earns the mark.

A1. What is internetworking? [Notes]

- Internetworking is the process of **connecting different networks** through intermediary devices such as routers or gateways.
- The interconnected networks are called an **internetwork**, or simply an internet.
- It is possible only if all the connected networks use the **same protocol stack**; the communication at the network layer is host-to-host.

A2. Write a short note on a router. [PYQ 2024-25 backlog]

- A router is a **three-layer device**: physical, data link and network.
- As a physical device it **regenerates bits**; as a data link device it checks physical addresses; as a network device it examines the **IP address** and forwards the packet on the best path.
- A **host** is the end system to which data are ultimately delivered; a **gateway** is the router that performs routing between two networks.

A3. Define routing. [Notes]

- Routing is the process of **selecting the shortest and most reliable path** over which to send data to its ultimate destination.
- It is done at the network layer, using the logical (IP) address.
- IP routing protocols distinguish **hosts** (end systems) from **gateways** (routers between networks).

A4. Find the class, net-id and host-id of the IP address 155.25.45.48. [PYQ 2023]

1. The first byte is 155, which lies in 128-191, so it is a **Class B** address.
2. For class B the first two bytes are the net-id: **155.25**.
3. The remaining two bytes are the host-id: **45.48**.
4. Default mask 255.255.0.0 (/16); the address is in the range 155.25.0.0 - 155.25.255.255.

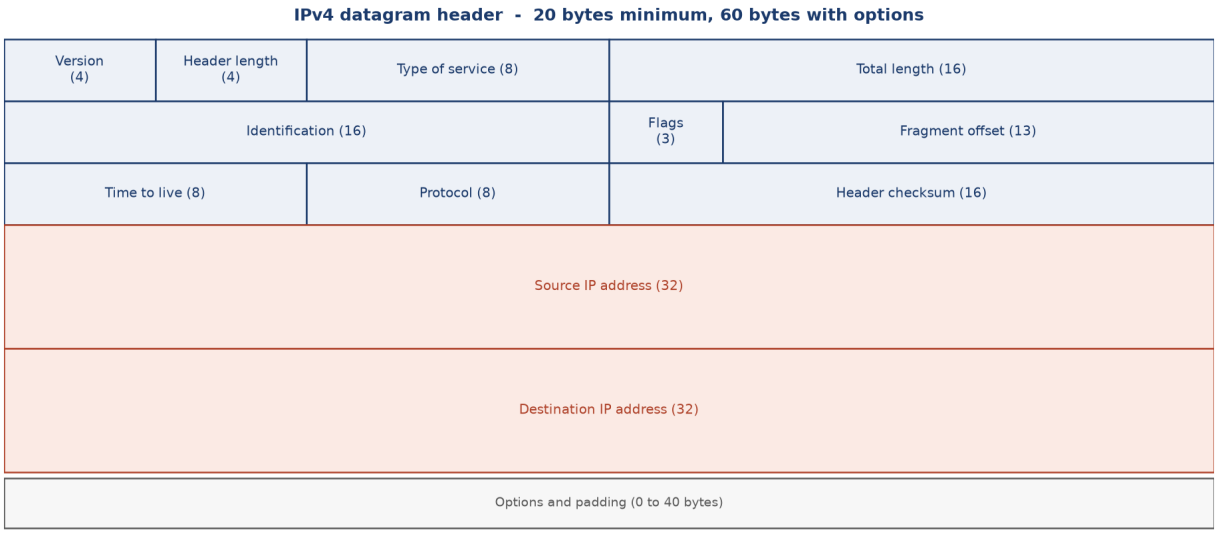
A5. Find the class of each address: 00000001 00001011 00001011 11101111, 11000001 10000011 00011011 11111111, 14.23.120.8, 252.5.15.111. [Notes]

1. First bit 0 - **Class A**.
2. First two bits 1 and third bit 0 - **Class C**.
3. First byte 14, in 1-126 - **Class A**.
4. First byte 252, in 240-255 - **Class E** (reserved).

A6. A subnet has the mask 255.255.255.192. Calculate the maximum number of hosts that can be configured. [PYQ 2023]

1. 192 = 11000000, so 2 host bits are left and the prefix is /26.
2. Total addresses per subnet = 2^6 = 64.
3. Usable hosts = 64 - 2 = **62** (network address and broadcast address are excluded).

A7. Draw a labelled diagram of the structure of the IPv4 datagram header. [PYQ 2024-25; Assignment 4]



- Header is **20 bytes minimum** and 60 bytes with options, in rows of 32 bits.
- Fields: version (4), header length (4), type of service (8), total length (16), identification (16), flags (3), fragment offset (13), time to live (8), protocol (8), header checksum (16), source address (32), destination address (32), options and padding.

A8. Outline the functions of ARP and RARP in an IPv4 network. [PYQ 2024-25]

- **ARP** (Address Resolution Protocol) - dynamic mapping from a **logical to a physical** address: given an IP address it finds the MAC address, so the packet can be handed to the data link layer.
- An ARP **request is broadcast**; the ARP **reply is unicast**.
- **RARP** (Reverse ARP) does the opposite: a host that knows only its physical address discovers its **IP address**, which is how a diskless workstation booted.
- RARP has been replaced by BOOTP and DHCP.

A9. Explain the importance of the ARP protocol in IP addressing. [PYQ 2022]

- Delivery needs **two levels of address**: the IP address to route the packet to the right network, and the MAC address to deliver the frame to the right NIC.
- ARP supplies the missing link - it maps a known IP address to its physical address dynamically, with no manual table.
- Without ARP the network layer could not hand a packet to the data link layer on a LAN.

A10. What is ICMP? Name its two message categories. [Notes]

- IP has no error-reporting or error-correcting mechanism and no way to query hosts; the **Internet Control Message Protocol** was designed to fill both gaps.
- **Error-reporting messages** go to the source host: destination unreachable (3), source quench (4), time exceeded (11), redirection (5), parameter problem (12).
- **Query messages** come in request and reply pairs: echo request (8) and echo reply (0) - the basis of ping - and timestamp request and reply (13, 14).

A11. What is IPv6? Give its advantages over IPv4. [PYQ 2022; Assignment 4]

- IPv6 uses **128-bit addresses** instead of 32, written as eight groups of four hexadecimal digits separated by colons - an address space of 3.4 x 10^38.
- **Built-in IPsec**: authentication and privacy, which IPv4 lacked.
- **Simplified header** with fixed fields and extension headers, so routers process packets faster.
- **No NAT needed**, so end-to-end communication is restored.
- Better support for **QoS** through the traffic class and flow label, and no broadcast - multicast and anycast instead.

A12. State the key differences between TCP and UDP. [PYQ 2022; PYQ 2024-25; Assignment 4]

Point	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Reliable - ACKs and retransmission	Unreliable - no ACK
Order	Ordered byte stream	No ordering
Header	20 bytes minimum	8 bytes
Flow and congestion control	Both	Neither
Speed and use	Slower; web, e-mail, file transfer	Faster; DNS, streaming, voice, SNMP

A13. Explain the TCP three-way handshake. [Notes]

1. The client sends **SYN** with its initial sequence number.
2. The server replies **SYN + ACK**, acknowledging that number and giving its own.
3. The client sends **ACK**; the connection is established and data can flow.
4. Termination is a four-step exchange of FIN and ACK in both directions.

A14. List the fields of the TCP segment header. [Assignment 4]

- Source port (16), destination port (16), sequence number (32), acknowledgement number (32).
- Header length (4), reserved (6), control flags (6: URG, ACK, PSH, RST, SYN, FIN), window size (16).
- Checksum (16), urgent pointer (16), options and padding, then data.
- Minimum header 20 bytes, maximum 60 bytes.

A15. Define congestion. [PYQ 2022; PYQ 2023]

- Congestion is a situation in which **too many packets** are present in a part of the subnet, so queues build up, delay rises and packets are dropped.
- It is a **global** condition of the network, not of one sender or one receiver.
- Congestion control is the set of techniques that **prevent** it before it happens or **remove** it after.

A16. Distinguish congestion control from flow control. [Notes]

- **Flow control** is between one sender and one receiver - it stops a fast sender from overrunning a slow receiver's buffer.
- **Congestion control** is a network-wide concern - it stops the total offered load from exceeding what the subnet can carry.
- A network can be congested even when every receiver is fast, and flow control alone does not prevent it.

A17. Differentiate open-loop and closed-loop congestion control. [Notes]

Point	Open loop (prevention)	Closed loop (removal)
When it acts	Before congestion happens	After congestion is detected
Mechanism	Good design and policies	Feedback from the network
Examples	Retransmission policy, window policy, admission control, traffic shaping	Backpressure, choke packet, implicit or explicit signalling

A18. What is backpressure? How does it differ from a choke packet? [Notes]

- **Backpressure** - a congested node **stops receiving** packets from its upstream node, which in turn stops receiving from above, so the warning travels hop by hop back to the source.
- **Choke packet** - the congested router sends a packet **directly to the source**, telling it to slow down; intermediate nodes are not warned.
- Backpressure is node-to-node and works only on virtual-circuit networks; the choke packet is router-to-source and works on datagram networks.

A19. Define flooding in a routing algorithm. [PYQ 2023]

- Flooding is a static routing technique in which every incoming packet is sent out on **every link except the one it arrived on**.
- It guarantees that a packet reaches the destination by the shortest path, so at least one copy arrives even if links fail.
- It generates a huge volume of traffic; controlled versions add a hop counter, a sequence number or selective flooding.

A20. Differentiate intra-domain and inter-domain routing. [PYQ 2023]

Point	Intra-domain (interior)	Inter-domain (exterior)
Scope	Inside one autonomous system	Between autonomous systems
Protocol type	IGP - RIP, OSPF	EGP - BGP
Goal	Shortest path, fast convergence	Policy, reachability, trust
Scale	Small to medium, one administration	Internet-wide, many administrations

A21. Explain the basic mechanism of distance-vector routing. [PYQ 2023]

1. Each router keeps a table of the **cost to every destination** and the first hop (next router) to reach it.
2. Periodically, and whenever its table changes, a router sends its **entire table** to its immediate neighbours.
3. A neighbour updates its own table by the Bellman-Ford rule: cost to X via neighbour N = cost to N + cost from N to X, keeping the smaller value.
4. Example: if A reaches B at cost 1 and B reports C at cost 2, then A sets the route to C at cost 3 via B.
5. Weakness: slow convergence and the **count-to-infinity** problem; RIP is the classic protocol.

A22. Name the ad hoc routing protocols in the syllabus and state one line about each. [Notes]

- **DSDV** (Destination-Sequenced Distance Vector) - **proactive**: every node keeps a full table with sequence numbers, which prevents loops.
- **DSR** (Dynamic Source Routing) - **reactive**: the sender discovers a route on demand and puts the whole route in each packet header, caching routes for reuse.
- **AODV** (Ad hoc On-demand Distance Vector) - **reactive**: route request broadcast and unicast reply, with hop-by-hop tables and sequence numbers instead of full source routes.

A23. Define classful and classless addressing. [Notes]

- **Classful**: the address space is divided into classes A to E and the boundary between net-id and host-id is **fixed** by the class - /8, /16 or /24. Much of the space was wasted, and it is now obsolete.
- **Classless**: the boundary is anywhere and is given by the prefix **/n** (CIDR). It allows VLSM and efficient use of addresses.
- In a block x.y.z.t/n, the **first address** is the network address and the **last** is the broadcast address.

A24. Find the first address of the block containing 205.16.37.39/28. [Notes]

1. Binary: 11001101 00010000 00100101 0010 0111.
2. Set the 32 - 28 = 4 rightmost bits to 0: 11001101 00010000 00100101 0010 0000.
3. **First address = 205.16.37.32**; the block runs to 205.16.37.47, so 14 hosts are usable.

A25. What is subnetting? Differentiate FLSM and VLSM. [Notes]

- Subnetting divides one physical network into several **logical subnets** using a longer mask, which improves efficiency, security and management.
- **FLSM** - fixed-length subnet mask: every subnet has the same mask and therefore the same number of hosts.
- **VLSM** - variable-length: each subnet gets a mask sized to its actual need, so addresses are not wasted.

A26. For 201.20.30.40/24, find the subnet mask, network ID, broadcast address and number of usable hosts. [Notes]

1. Default mask = **255.255.255.0** (/24), since 201 is class C.
2. Network ID = **201.20.30.0**.
3. Broadcast address = **201.20.30.255**.
4. Hosts per subnet = 2^8 - 2 = **254**.
5. Bits borrowed = 24 - 24 = 0, so there is 1 subnet.

A27. For 183.42.0.0/22, find the class, the default mask, the custom mask, the number of subnets and the number of hosts per subnet. [PYQ 2024-25]

1. First byte 183 - **Class B**; default mask **255.255.0.0** (/16).
2. Custom mask for /22 = **255.255.252.0**.
3. Borrowed bits = 22 - 16 = 6, so subnets = 2^6 = **64**.
4. Host bits = 32 - 22 = 10, so addresses = 1024 and usable hosts = **1022**.
5. Block size in the third octet = 4: subnets are 183.42.0.0, 183.42.4.0, 183.42.8.0 ...; broadcasts 183.42.3.255, 183.42.7.255, 183.42.11.255 ...

A28. For 194.32.47.40/24, find the network ID, the first and last host, the broadcast address, the subnets and the mask. [PYQ 2022]

1. Mask = **255.255.255.0** (/24), class C.
2. Network ID = **194.32.47.0**; broadcast = **194.32.47.255**.
3. First host = **194.32.47.1**; last host = **194.32.47.254**.
4. Subnets = 1 (no bits borrowed), valid hosts = **254**.

A29. For 180.20.10.20/20, find the class, the default mask and the custom mask. [Assignment 4]

1. First byte 180 - **Class B**; default mask **255.255.0.0**.
2. Custom mask for /20 = **255.255.240.0**.
3. Borrowed bits = 4, so 16 subnets; host bits = 12, so 4096 addresses and **4094** usable hosts per subnet.

A30. What is a port address? Give the well-known ports you must remember. [Notes]

- A port address is a **16-bit** number that identifies the process on the destination host; the transport layer uses it for process-to-process delivery.
- **Ports to memorise**: FTP 20 (data) and 21 (control), Telnet 23, SMTP 25, DNS 53, HTTP 80, POP3 110, IMAP 143, HTTPS 443.

A31. List the fields of the UDP header. [Notes]

- **Source port** (16 bits) and **destination port** (16 bits).
- **Total length** (16 bits) of the user datagram.
- **Checksum** (16 bits) for error detection.
- Only 8 bytes in all - which is why UDP is cheap.

A32. What is the size of the IPv4 address space, and what is a subnet mask? [Notes]

- An IPv4 address is 32 bits, so the address space is 2^32 = **4,294,967,296** addresses.
- A **mask** separates the network part from the host part: a 1 bit means network, a 0 bit means host.
- Default masks: class A 255.0.0.0, class B 255.255.0.0, class C 255.255.255.0.

A33. What is a datagram? Is IP reliable? [Notes]

- A datagram is the packet in which IP transports data; each is handled **independently** of all others.
- Datagrams may take different routes, arrive out of order, or be dropped.
- IP is therefore **connectionless and unreliable** (best effort); reliability is added by TCP above it.

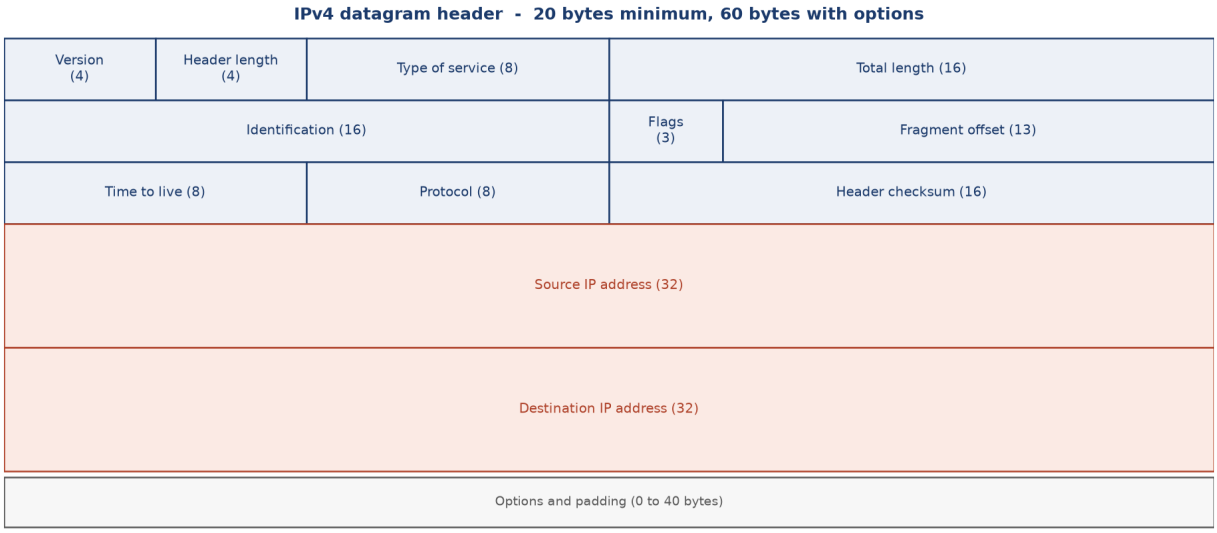
A34. What is the purpose of the time-to-live field? [Notes]

- Every router decrements TTL by 1; when it reaches 0 the router **discards the datagram** and sends an ICMP time-exceeded message.
- It prevents a datagram from circulating forever when routing tables are wrong.
- It is also the mechanism traceroute exploits.

Section B | 7-mark questions

Write the heading points first, then expand each one. The bullet structure below is the marking scheme: one bullet is worth roughly half a mark to a mark.

B1. Draw the schematic diagram of the IPv4 header, explain its fields, and give the advantages of IPv6 over IPv4. [PYQ 2022; Assignment 4]



- **Version** (4 bits) - IP version, 4 here.
- **Header length** (4 bits) - length of the header in 4-byte words, 5 to 15, because options make it variable.
- **Type of service** (8 bits) - precedence, delay, throughput and reliability hints.
- **Total length** (16 bits) - header plus data, so a datagram is at most 65,535 bytes.
- **Identification** (16 bits) - identifies the datagram so fragments can be reassembled.
- **Flags** (3 bits) - do not fragment, more fragments.
- **Fragment offset** (13 bits) - position of this fragment in the original datagram, in 8-byte units.
- **Time to live** (8 bits) - hop limit, decremented by each router.
- **Protocol** (8 bits) - which upper-layer protocol owns the payload: TCP (6), UDP (17), ICMP (1).
- **Header checksum** (16 bits) - covers the header only, so it is recomputed at every router.
- **Source and destination address** (32 bits each).
- **Options and padding** - up to 40 bytes, padded to a 4-byte boundary.

IPv6 advantages. A 128-bit address space removes exhaustion; IPsec is built in for authentication and privacy; the header is simplified and fixed-length with extension headers, so routers process it faster; NAT is unnecessary, restoring true end-to-end connectivity; the traffic class and flow label support quality of service; and broadcast is replaced by multicast and anycast.

B2. Compare IPv4 and IPv6. [Assignment 4]

Point	IPv4	IPv6
Address size	32 bits	128 bits
Notation	Dotted decimal	8 groups of 4 hex digits, colon separated

Point	IPv4	IPv6
Header	Variable, 20-60 bytes, 13 fields	Fixed 40 bytes plus extension headers
Checksum	Header checksum present	Removed - left to upper layers
Fragmentation	Done by routers and hosts	Done by the source only
Security	IPsec optional add-on	IPsec mandatory part of the standard
Address configuration	Manual or DHCP; NAT common	Stateless autoconfiguration; no NAT needed
Broadcast	Supported	Replaced by multicast and anycast

B3. Explain the TCP segment format with a neat diagram, and compare TCP with UDP. [Assignment 4]

- **Source and destination port** (16 bits each) - identify the two processes.
- **Sequence number** (32 bits) - position of the first data byte in the stream; the initial value is random.
- **Acknowledgement number** (32 bits) - the next byte expected; valid only when the ACK flag is set.
- **Header length** (4 bits) - 5 to 15 words, so 20 to 60 bytes.
- **Reserved** (6 bits) and **control flags** (6 bits: URG, ACK, PSH, RST, SYN, FIN).
- **Window size** (16 bits) - how much the receiver can accept, which is the flow-control mechanism.
- **Checksum** (16 bits) - covers header, data and a pseudo-header.
- **Urgent pointer** (16 bits) and **options and padding**.

Point	TCP	UDP
Connection	Connection-oriented	Connectionless
Header	20 to 60 bytes	8 bytes
Reliability	ACKs, retransmission, ordering	None
Flow and congestion control	Window and slow start	None
Speed and overhead	Slower, high overhead	Fast, minimal overhead
Typical use	HTTP, FTP, SMTP, Telnet	DNS, DHCP, streaming, VoIP

B4. For the given IP address 183.42.0.0/22, find the class, default and custom mask, number of subnets, number of hosts per subnet, valid subnets and the broadcast address of each subnet. [PYQ 2024-25]

1. **Class**: first byte 183, so class B.
2. **Default mask**: 255.255.0.0 (/16).
3. **Custom mask**: 255.255.252.0 (/22).
4. **Borrowed bits** = 22 - 16 = 6, so number of subnets = 2^6 = **64**.
5. **Host bits** = 32 - 22 = 10, so addresses per subnet = 1024 and usable hosts = **1022**.
6. **Block size** in the third octet = 256 - 252 = 4.
7. **Valid subnets**: 183.42.0.0, 183.42.4.0, 183.42.8.0, 183.42.12.0 ... 183.42.252.0.
8. **Broadcast of each subnet**: 183.42.3.255, 183.42.7.255, 183.42.11.255 ... 183.42.255.255.

Routine to memorise: class, then default mask, then borrowed bits = prefix - default prefix, then subnets = 2^borrowed, then host bits = 32 - prefix, then hosts = 2^host bits - 2, then block size = 256 - mask octet.

B5. For the IP address 194.32.47.40/24, find the network ID, the first and last host IDs, the broadcast address, the subnets, the valid hosts and the subnet mask. [PYQ 2022]

1. **Mask**: 255.255.255.0 (/24), a class C address.
2. **Network ID**: 194.32.47.0.
3. **First host**: 194.32.47.1; **last host**: 194.32.47.254.
4. **Broadcast**: 194.32.47.255.
5. **Subnets**: 1, because no bits are borrowed from the host part.
6. **Valid hosts**: 254 (256 - 2).

If the same block were subnetted to /26, the mask would be 255.255.255.192 and four subnets of 62 hosts would appear: 194.32.47.0, .64, .128 and .192, with broadcasts .63, .127, .191 and .255.

B6. For the IPv4 address 180.20.10.20/20, determine the class, the default and custom masks, the number of subnets, the usable hosts, the valid subnets and the network and broadcast of the subnet containing the address. [Assignment 4]

1. **Class**: B (first byte 180).
2. **Default mask**: 255.255.0.0 (/16); **custom mask**: 255.255.240.0 (/20).
3. **Borrowed bits** = 4, so subnets = 2^4 = **16**.
4. **Host bits** = 12, so usable hosts per subnet = 2^12 - 2 = **4094**.
5. **Block size** = 256 - 240 = 16, so the subnets are 180.20.0.0, 180.20.16.0, 180.20.32.0 ... 180.20.240.0.
6. 10 falls in the first block, so 180.20.10.20 belongs to subnet **180.20.0.0**.
7. Its broadcast is **180.20.15.255**, so its usable range is 180.20.0.1 to 180.20.15.254.

B7. Define congestion. Explain the principle and the prevention policies of congestion control. [PYQ 2022; PYQ 2023; PYQ 2024-25; Assignment 4]

Definition. Congestion occurs when the number of packets injected into the network exceeds what the network can carry, so queues grow, delay rises and packets are discarded. It is a global condition, unlike flow control, which is local between one sender and one receiver.

- **Open-loop (prevention)** policies try to stop congestion before it happens, by designing the system well:
- **Retransmission policy** - a sender must not retransmit too eagerly; a careless retransmission policy adds load exactly when the network is already full.
- **Window policy** - a selective-repeat window puts less load on the network than go-back-N.
- **Acknowledgement policy** - fewer ACKs mean less traffic; piggybacking helps.
- **Discarding policy** - drop the least damaging packets (for example the less important ones in a video stream) rather than any packet.
- **Admission policy** - refuse new flows when the network is near capacity.
- **Traffic shaping** - leaky bucket and token bucket smooth a bursty source into a steady rate.
- **Closed-loop (removal)** policies act after congestion is detected: **backpressure** (a congested node stops taking packets from its upstream neighbour), **choke packet** (a router warns the source directly), and **implicit or explicit signalling** (delay or marked packets tell the source to slow down).

One-line answer for the exam: prevent by good policy at the sender and at the admission gate; remove by feeding information back from the congested node to the source.

B8. Differentiate intra-domain and inter-domain routing and explain distance-vector routing with an example. [PYQ 2023]

Point	Intra-domain	Inter-domain
Scope	Within one autonomous system	Between autonomous systems
Protocols	RIP, OSPF (IGP)	BGP (EGP)
Objective	Least cost path, quick convergence	Policy and reachability, not just cost
Administration	Single	Many, often competing

1. Distance-vector routing is an intra-domain method used by RIP.
2. Each router's table has one row per destination: the cost and the next hop.
3. Every router periodically sends its **whole table** to its direct neighbours - hence distance vector.
4. A router recomputes each entry by Bellman-Ford: new cost to X via neighbour N = cost(N) + table of N for X. Keep the smallest.
5. **Example**: A is linked to B at cost 1 and to C at cost 2; B reports C at cost 2. A then reaches C at cost 1 + 2 = 3 via B, but keeps its direct route at cost 2.
6. Problems: slow convergence and count-to-infinity, fixed by a split horizon, a hop limit (15 in RIP) or poison reverse.

Compare: distance vector knows only its neighbours' views; link-state (OSPF) floods the whole topology and runs Dijkstra on it.

B9. Explain the network layer protocols ARP, RARP, ICMP and IGMP. [Notes]

- **ARP** - maps a known IP address to a MAC address by broadcasting a request and receiving a unicast reply; it is what lets a packet leave the host on a LAN.
- **RARP** - the reverse: a host with only a physical address asks for its IP address; now replaced by DHCP.
- **ICMP** - reports errors back to the source (destination unreachable, time exceeded, redirection, parameter problem, source quench) and carries query pairs such as echo request and reply for ping and traceroute.
- **IGMP** - manages multicast group membership so that one stream can reach many receivers without duplication.

B10. Explain classful addressing: classes, net-id and host-id, default masks and its drawbacks. [Notes]

Class	First byte	Net-id / host-id	Default mask	Hosts per network
A	1-126	1 byte / 3 bytes	255.0.0.0	16,777,214
B	128-191	2 bytes / 2 bytes	255.255.0.0	65,534
C	192-223	3 bytes / 1 byte	255.255.255.0	254
D	224-239	Multicast	Not applicable	-
E	240-255	Reserved	Not applicable	-

- **Drawbacks**: a large part of the space was wasted, because an organisation that needed 300 addresses had to take a class B of 65,534.
- Classful addressing is almost obsolete; classless addressing with CIDR (/n) replaced it, and it supports VLSM.

B11. Explain TCP connection management: the three-way handshake, termination and flow control. [Notes]

1. **Handshake**: client sends SYN with sequence number x; server replies SYN + ACK acknowledging x + 1 with its own sequence number y; client sends ACK for y + 1. The connection is now open.
2. **Data transfer**: bytes are numbered from a random initial value, acknowledgements are cumulative, and lost segments are retransmitted after a timeout or three duplicate ACKs.
3. **Termination**: one side sends FIN, the other ACKs, then the other side sends its own FIN, which is ACKed - four steps because TCP is full-duplex.
4. **Flow control**: the receiver advertises a window size; the sender may have at most that much unacknowledged data outstanding.
5. **Congestion control**: slow start, congestion avoidance, fast retransmit and fast recovery adjust the effective window to the network's capacity.

B12. Explain the ad hoc routing protocols DSDV, DSR and AODV. [Notes]

- **DSDV - proactive, table driven**: every node keeps a complete routing table with a **destination sequence number**, which is used to choose the freshest route and to prevent loops. Updates are periodic and incremental. It routes immediately but spends bandwidth on updates even when nothing is sent.
- **DSR - reactive, source routing**: on demand, the source floods a route request; the reply carries the path, and the source puts the **complete route in each packet header**. Routes are cached. Overhead is low when traffic is sparse, but the header grows with the path length.
- **AODV - reactive, hop by hop**: route request broadcast and unicast reply, as in DSR, but each node keeps only the **next hop** in its table and uses sequence numbers plus a route lifetime, so headers stay small and stale routes expire.
- **Choosing**: proactive for steady traffic, reactive for occasional traffic in a fast-changing topology - which is why AODV and DSR dominate in mobile ad hoc networks.